Hyperconical gravity applied to the renormalization scale

One cone read as time and as scale: hyperconical modified gravity, a scale-dependent vacuum and one galactic acceleration scale

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Abstract

Hyperconical modified gravity (HMG) describes our universe as a flat slice of a five-dimensional hypercone whose radius grows with the age of the universe, $R = ct$. Because radius and age are tied, the cone can be read as time, each slice the universe at one age, or as scale, each slice one layer of the structures that exist together today. A framework that treats scale as a coordinate S of a warped geometry, in which couplings run and the vacuum is a medium that changes with S, makes the same move for every structure. Both are reinterpretations of known physics on a scale direction, built independently; we read the cone as scale, a postulate stated as such, and place both readings on one sheet (S,t): HMG's universe is its edge, a cut at one time gives the layers, and a cut at one layer gives its history. Across S the four interactions are one field read at different scales, in four layers (weak, strong, electromagnetic, gravity), each with its own vacuum at its own scale; HMG's slice formula gives that vacuum, which falls as gravity's coupling, by 10^{120} from the floor to our layer, and the stack of media below each layer gives the gauge couplings ($\alpha_s = 0.224$ and 0.313 at the bottom-quark and τ scales, $1/\alpha_{\rm em} = 136.7$ at the electron). The vacuum changes sign at the two edges of the observed range, the floor and our layer, and the axis continues past both. Where the edge meets our layer, with nothing fitted, HMG's vacuum is $0.694\,\rho_{\rm crit}$ against 0.697 ± 0.004 measured; its Jeans rate gives the galactic scale $g_\dagger = 1.06 \times 10^{-10}\,{\rm m\,s^{-2}}$, which is HMG's $a = 2c/(\gamma_0 t)$ with $\gamma_0 = 12.3$; isolated KiDS-1000 lenses read it directly, and the excess around early-type lenses is the pull of neighbours. Through time, every interaction layer holds still, from today back to the CMB, and riding the edge is excluded at about 1900σ; only our vacuum layer drifts, slowly ($p = 0.011 \pm 0.005$). The galactic scale is therefore predicted to stay constant to within 1% out to $z = 2$, and the vacuum's equation of state never to cross -1. Today the edge crosses the layer of the vacuum's own clock, near the epoch at which observers appear.

Keywords: hyperconical modified gravity; renormalization scale; vacuum energy; radial acceleration relation; dark energy; cosmological constant problem

1 Introduction

Hyperconical modified gravity (HMG) [1, 2, 3, 4] models the universe as a hypercone embedded in flat five-dimensional space. Our universe is a slice whose radius grows as $R = ct$, so the expansion is linear, $H = 1/t$, and the background density is the critical one, $\rho_0 = 3/(8\pi G t^2)$. Projected onto the observer's frame, this geometry looks like a flat universe with $\Omega_\Lambda = 0.6937$, obtained by matching the projected expansion to the standard one at second order in redshift [1]. The same projection gives a residual acceleration $a = 2c/(\gamma_0 t)$, whose size is set by a projection angle; matching the galactic scale requires

$\gamma_0 \approx 13 \pm 3$ [3], and fits to rotation curves give angles $0.460 \pm 0.002\,\pi$ and $0.468 \pm 0.003\,\pi$ [2]. The angle depends on the local balance between a structure's own pull and the Hubble flow, which gives HMG an environment model for lenses and clusters [4, 5].

A different framework [6] treats the renormalization scale as a coordinate,

$$S = \ln \frac{R}{R_{\text{floor}}}, \qquad R_{\text{floor}} = \sqrt{2}\,\ell_P,$$

where R is a structure's radius and R_{floor} is where the quantum line $R = \hbar/Mc$ meets the black-hole line $R = 2GM/c^2$. In a warped geometry with this coordinate, moving along S is renormalization-group running, and the vacuum is a medium whose curvature changes with S. The galactic acceleration scale is then the Jeans rate of that medium [7].

On HMG's cone a slice's radius and its age are tied, $R = ct$: towards the tip the slices are both smaller and earlier. HMG reads the cone's axis as time, so its slices are the universe at successive ages. The axis can also be read as scale, so that its slices are the layers of structure, from the floor to the universe, that exist together today. Both frameworks are reinterpretations: neither adds new particles or fields, and each reads known physics, the expansion and the galactic scale in HMG, the running couplings, the vacuum-energy hierarchy and the radial acceleration relation in [6, 7], as consequences of a scale direction. HMG builds that direction for the universe as a whole; [6] builds it for every structure. The step that joins them is a reading, not a derivation: we take the cone's axis as scale and give each layer of structure the clock $t = R/c$, and we state this as the postulate of the paper. This paper places both readings on one sheet with coordinates (S,t) and shows that they are three cuts through it (Section 2). HMG's universe is one path on the sheet, its edge; the layers and the interactions are a cut across S at one time (Sections 3 and 4); the edge meets our layer today, where every number is measured (Sections 5 and 6); a cut along time at every layer decides which reading carries the vacuum and the interactions (Section 7); and the edge with its active layer, in the standard history, places the early universe (Section 8). Every test uses public data, with the pass rule fixed before running and nothing fitted to the data tested.

2 One sheet, three cuts

The geometry of the scale axis,

$$ds^2 = e^{2A(S)}\left(-dt^2 + a(t)^2 d\vec{x}^2\right) + dS^2,$$

has two coordinates besides space, time t and scale S, with every slice of fixed S flat in space, as measured for our own ($\Omega_K = 0.0005 \pm 0.0020$ [8]). It is read holographically, so space keeps three dimensions. In the warped bulk S is proportional to $\ln\mu$, and d/dS is the renormalization-group derivative: a coupling placed along S runs as the Standard Model says [6].

Every point of the (S,t) sheet is one scale at one moment (Figure 1). Three cuts through it carry the physics of this paper.

The edge. The universe's own size is a path on the sheet, the Hubble radius c/H. In the late universe it is HMG's slice, $R = ct$,

$$S_{\text{u}}(t) = \ln \frac{ct}{R_{\text{floor}}},$$

at $S = 139.90$ today; from $z = 2$ to today ct and the standard c/H agree within 0.01 e-folds. At early times the edge follows the standard history (Section 8). Everything below the edge exists at that time: protons existed when the universe was a billion years old, without being its largest layer. HMG's cosmology is the physics read along this path.

Across S at one time. At fixed t the sheet is a stack of layers, each with its own size and clock R/c. HMG's slice formula, evaluated with each layer's radius, gives each layer its vacuum (Section 4); the medium of each layer, made of the structures just below it, gives the gauge couplings (Section 3). This is HMG's cone read as scale, the postulate of the paper.

Along time at one S. At fixed S the sheet shows whether a layer changes as the edge moves. If the vacuum seen as dark energy were carried by the edge, its layer would move with it and Ω_Λ would be the same at every age; if it belongs to its place on S, the layer holds still and the edge moves through it. Section 7 measures which, for our layer and for every layer that carries an interaction.

The three cuts give three maps. The first follows the edge: the universe growing, slice by slice, with the layer that is active at each moment (Section 8). The second is one frame: every scale at one time, with the interactions across it (Sections 3 and 4). The third is the film: every layer through time, frame by frame (Section 7). Each frame of the film is a map of the second kind, bounded by the edge of the first.

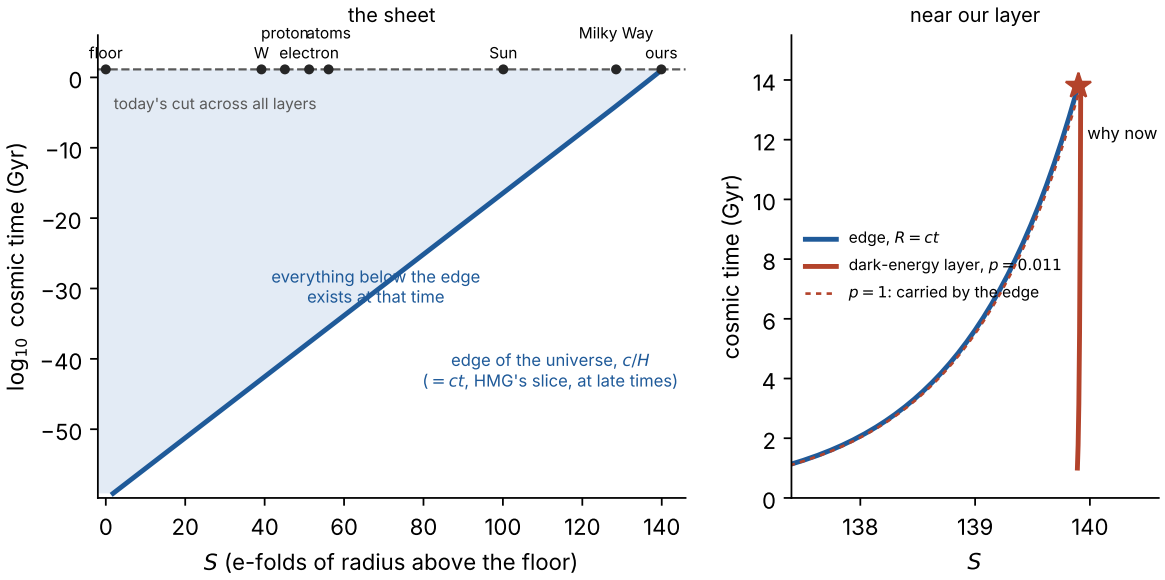


Figure 1: The (S,t) sheet. Left: the edge of the universe, c/H, equal to ct (HMG's slice) at late times, and today's cut through the layers of Table 1. Right: the last 13 Gyr near our layer. The layer of the vacuum seen as dark energy, from the measured $p = 0.011$ (Section 7), holds still while the edge moves through it; $p = 1$ would carry it along the edge. They cross today (Section 8).

3 Across S today: four force layers

Across S at one time the four interactions are one field read at different scales [6]. Layers of different S coexist in the same place: going down S at a given place goes inward, toward the centre of mass, through smaller and smaller structure. The vacuum medium is tight at small S and flat at large S, and the structures of each layer are the medium of the layer above; the layers push on each other along the axis, the smaller up against the larger and the larger down on the smaller. Read along S this gives four force layers, weak, strong, electromagnetic and gravity (Table 1). Each layer has its own vacuum energy at its own scale, the dark energy of that size (Section 4); the dark energy we measure is the vacuum of our layer. Matter relaxing the medium around it is a separate, local effect, which at the galactic layer appears as unseen mass (Section 5).

Figure 2 shows the measured couplings across S, run from their values at the Z mass. Gravity's coupling for a quantum of each layer's scale, $(\ell_P/R)^2$, falls from 0.5 at the floor to 10^{-45} at the electron and 10^{-122} at our layer; the gauge couplings change by a fixed amount per e-fold.

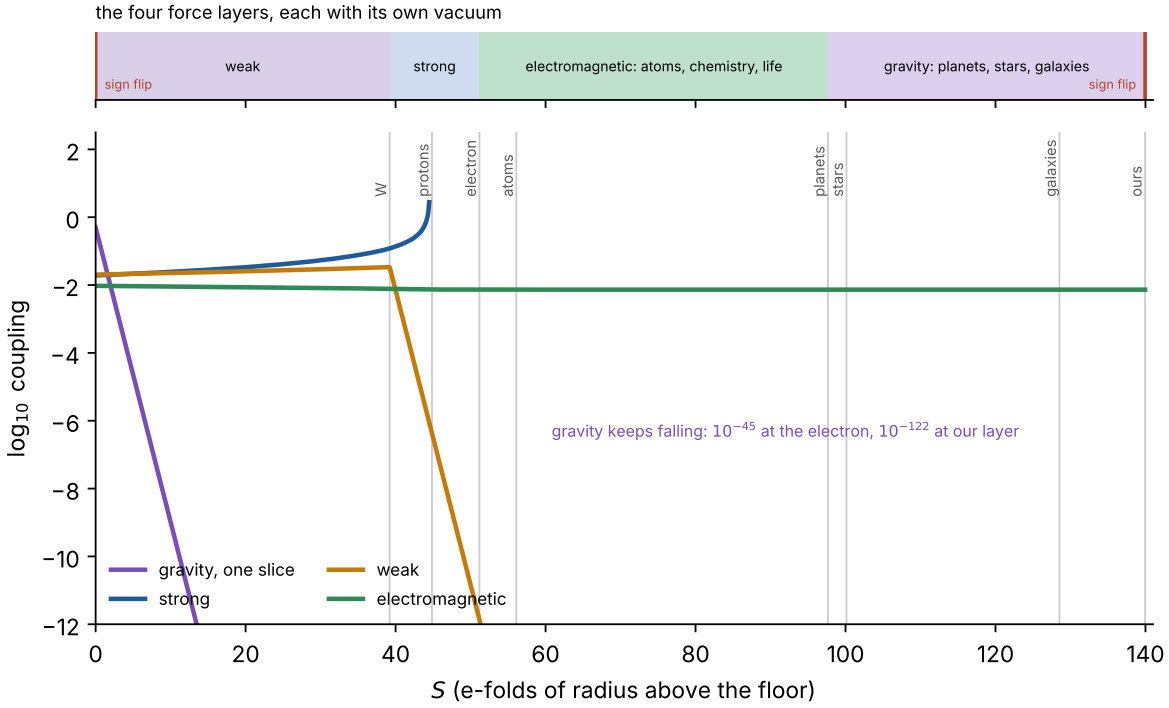


Figure 2: One frame of the sheet: the couplings from the floor to our layer, today. Gravity, the projection of one slice, is $(\mu/M_P)^2$; the gauge couplings are the stack of media below each layer, run from their values at the Z mass (one loop above it; two loops for the strong medium below it). The weak interaction is short-range below the W mass, the strong medium binds into hadrons at confinement, and electromagnetism stays at $1/137$ from the electron to our layer. Top: the four force layers, each with its own vacuum, and the sign flips at the floor and at our layer (Section 4).

The couplings along S: the slice and the stack. The measured couplings change with S in two different ways, and the two map onto the two halves of the picture. Gravity's coupling for a quantum of the layer's scale, $(\ell_P/R)^2$, changes as a power of the radius, that is exponentially in S; this is the dependence of a single slice, and Section 4 shows that HMG's slice vacuum follows it exactly. The gauge couplings instead change logarithmically with scale, linearly in S: their inverse grows or falls by a fixed amount per e-fold. That is the dependence of a sum over layers. A charge measured at a given S is screened, or anti-screened, by the medium of every layer between the measurement and the slice, each e-fold of medium contributing a fixed amount set by the species it contains:

$$\frac{d\,\alpha_i^{-1}}{dS} = \frac{b_i(S)}{2\pi}, \tag{1}$$

where $b_i(S)$ counts the species active in the medium at that place on the axis, with their charges and spins. This is vacuum polarisation, read on the axis as the medium below acting on the layer above. The two signs correspond to HMG's two branches: a medium on the neighbourhood side of balance weakens the coupling at short distance, as the gluon medium does for the strong interaction, and a medium on the Hubble side strengthens it, as charged matter does for electromagnetism. Gravity is the projection of one slice; the gauge interactions are the projection of the stack of slices below. Neither half gives the

S	Radius	What happens	Force layer
0	2.3×10^{-35} m	The floor: quantum and black-hole lines meet; lower edge of the observed range, where the vacuum changes sign; the axis continues below	–
39.2	2.5×10^{-18} m	The weak interaction becomes short-range (W mass)	Weak
44.9	6.9×10^{-16} m	$\alpha_s = 1$: confinement; the proton sits at 45.1	Strong
51.2	3.9×10^{-13} m	Electromagnetism freezes at $1/137$ (electron mass)	Electromagnetic
56.1	5.3×10^{-11} m	Atoms and chemistry	Electromagnetic
58–98		Earth's life window: chemistry and an environment that sustains life; Earth's living systems lie at 60.6–83.6	Electromagnetic
97.7	5.9×10^{7} m	Largest cold body, $\sqrt{\alpha/\alpha_G}\,a_0$	Gravity
100.1	7.0×10^{8} m	Stars, mass $\alpha_G^{-3/2} m_p$	Gravity
128.5	49 kpc	Galaxies: matter relaxes the medium locally; the galactic scale appears	Gravity
131.5	1 Mpc	Clusters: hot plasma acts as part of the medium	Gravity
139.9	c/H_0	Our layer: its vacuum is the dark energy we measure; upper edge of the observed range, where the vacuum changes sign; the axis continues above	Gravity

Table 1: The four force layers along S, with the two edges of the observed range. Particle rows from the running of Figure 2; structure rows from [6], which agree with standard scaling arguments [9, 10].

interactions alone: a slice's geometry carries no species, and the medium carries no geometry.

Summed layer by layer from one baseline per coupling at the Z mass, $\alpha_s(M_Z) = 0.1180$ and $1/\alpha_{\rm em}(M_Z) = 127.95$, with the species active in each stretch of S (two loops for the strong medium, one loop for the charged medium), the stack gives the values in Table 2. The stack also marks where each interaction stops changing: below confinement ($S = 44.9$) the strong medium is bound into hadrons, and below the electron ($S = 51.2$), the lightest charged species, electromagnetism holds at its low-energy value through every larger layer up to ours.

Quantity	S	Stack	Measured
α_s, bottom-quark scale	42.2	0.224	0.225 ± 0.010
α_s, τ scale	43.0	0.313	0.325 ± 0.015
$1/\alpha_{\rm em}$, electron	51.2	136.7	137.036

Table 2: The stack of Eq. (1) against measurement, from one baseline per coupling at the Z mass. The electromagnetic value includes the leptons and the quarks above confinement; the remaining 0.3% is the hadronic contribution, which is taken from data.

4 Across S today: the vacuum of every layer

Give each layer the radius R of its structures and the clock $t = R/c$, so that HMG's slice formula is read with the cone's axis as scale. With HMG's projected fraction $\Omega_\Lambda = 0.6937$, which has no scale in it, the layer's vacuum and its Jeans rate are

$$\rho_{\rm vac}(S) = \Omega_\Lambda \frac{3}{8\pi G t^2} = 3\,\Omega_\Lambda M_p^2\,\mu^2(S), \qquad \omega_J = \sqrt{4\pi G\rho_{\rm vac}} = \sqrt{\tfrac{3\Omega_\Lambda}{2}}\,\frac{c}{R} = 1.02\,\frac{c}{R}. \tag{2}$$

On the edge, where the layer is the universe itself, this is HMG's own vacuum; below the edge it is the vacuum of each layer of structure, the dark energy of that size.

The vacuum curve is gravity's coupling. Measured against its value at the Planck length, the slice vacuum at radius R is $(\ell_P/R)^2$, which is gravity's coupling $(\mu/M_P)^2$ for a quantum of the layer's scale $\mu = \hbar c/R$. The vacuum curve and gravity's running are one line on the axis (Figure 3), and between the floor and our layer the vacuum falls by $(M_P/H_0)^2 \approx 10^{120}$. The quantum-field estimates of the vacuum and the observed cosmological constant are values of this one function at different places on S [6].

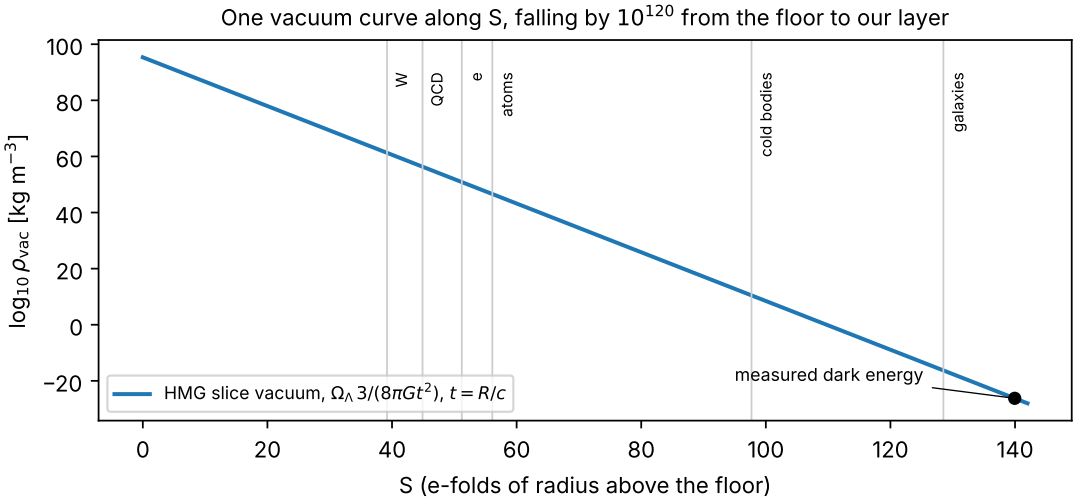


Figure 3: The vacuum of Eq. (2) across S today, with the measured dark energy at our layer.

Layer	S	Radius	Vacuum (kg m^{-3})	$c\,\omega_J/2\pi$ (m s^{-2})
Floor	0	2.3×10^{-35} m	2×10^{95}	6×10^{50}
W mass	39.2	2.5×10^{-18} m	2×10^{61}	6×10^{33}
Confinement	44.9	6.9×10^{-16} m	2×10^{56}	2×10^{31}
Electron	51.2	3.9×10^{-13} m	7×10^{50}	4×10^{28}
Bohr radius	56.1	5.3×10^{-11} m	4×10^{46}	3×10^{26}
Largest cold body	97.7	5.9×10^{7} m	3×10^{10}	2×10^{8}
Milky Way with halo	128.5	49 kpc	5×10^{-17}	1×10^{-5}
Our layer	139.9	c/H_0	5.9×10^{-27}	1.06×10^{-10}

Table 3: HMG's slice formulas on the layers of Table 1, read with the cone's axis as scale.

Two parts of a layer's vacuum. In the warped geometry a slice's vacuum is [11, 6]

$$\Lambda_{\text{eff}} = \tfrac{1}{2}\Lambda_5 + 3A'^2.$$

The second term is the bending of the geometry along S produced by the energy the slice carries; this is the part HMG obtains as projection, and with no bulk term it gives the full critical density, HMG's ρ_0. The first term is the medium below the slice, with $\Lambda_5 < 0$, which pulls the bending back. Our slice sits just above the point where its vacuum changes sign: the medium removes 30% of the bending, the share given by HMG's projected Ω_Λ. The part of the energy below that is carried inside bound structures gravitates as matter, consistent with the vacuum at each scale matching the bound structures just below it (within 0.30 dex at the QCD scale and 0.36 dex at the cosmic scale [6]).

The sign flip at both edges. The vacuum switches between repulsive and attractive at the two edges of the observed range: at the floor, where the quantum and black-hole lines meet, and at our layer, where the universe lies on its own black-hole line. Neither edge ends the axis; S continues below the floor and above our layer [6], so the observed range is one window on it.

5 Where the edge meets our layer

Today the edge sits at our layer, $S = 139.9$, and the two readings of the cone describe the same slice. Every quantity in this section is measured there.

The vacuum. At our layer ($R = c/H_0$) Eq. (2) gives $0.6937\,\rho_{\rm crit}$. Fitting Pantheon+ supernovae [13], DESI DR2 BAO [14] and Planck distance priors [15] together, in the family of Section 7, gives $\Omega_{\rm vac} = 0.697 \pm 0.004$; HMG's value lies at $\Delta\chi^2 = 0.94$, or 1.0σ.

The galactic scale. The Jeans rate of the slice's vacuum sets the galactic acceleration scale [7],

$$g_\dagger = \frac{c\,\omega_J}{2\pi} = \frac{c\sqrt{4\pi G\rho_{\rm vac}}}{2\pi} = 1.06 \times 10^{-10}\ {\rm m\,s^{-2}}. \tag{3}$$

In HMG's notation this is $a = 2c/(\gamma_0 t)$ with

$$\gamma_0 = \frac{4\pi}{\sqrt{3\Omega_\Lambda/2}} = 12.3,$$

inside the 13 ± 3 required to match the galactic scale [3]. Solving $\gamma_0 = \gamma/\cos\gamma$ gives the projection angle $\gamma = 0.4624\,\pi$, between the two fitted values $0.460 \pm 0.002\,\pi$ and $0.468 \pm 0.003\,\pi$ [2]. The scale lies within the measured $1.20 \pm 0.02\,({\rm random}) \pm 0.24\,({\rm syst}) \times 10^{-10}\ {\rm m\,s^{-2}}$ [16].

The shape. Matter relaxes the medium wherever its own gravitational rate beats the medium's clock. Reading the relaxed vacuum as a mode in its ground state, thermally excited by a bath at the vacuum's Jeans temperature, and requiring invariance under shifts along S far from matter, gives [7]

$$g = \frac{g_N}{1 - e^{-\sqrt{g_N/g_\dagger}}}, \tag{4}$$

the observed radial acceleration relation [16]. Far from matter it gives flat rotation curves and $v^4 = GMg_\dagger$.

Dataset	Result, nothing fitted	Pass rule
SPARC, 153 rotation curves [17]	$\Delta{\rm BIC} = -7.8$ against a free scale (free fit 1.054×10^{-10})	< 10
SPARC, form of the relation	Eq. (4) preferred over $g = \sqrt{g_N^2 + g_N a}$ by $\Delta{\rm BIC} = 113$	> 10
SPARC baryonic Tully–Fisher, 129 galaxies	observed $-$ predicted $= -0.02 \pm 0.02$ dex	$< 2\sigma$, < 0.1 dex
Isolated-lens circular velocities [18]	$\Delta\chi^2 = +3.2$	< 4
KiDS-1000 lensing RAR, late-type and blue lenses [20]	$\Delta\chi^2 = +1.2$	< 4

Table 4: Tests of the galactic scale and the form of the relation on our slice.

One clock from the Solar System to galaxies. Every structure inside our slice reads the same clock, Eq. (3). With the form of Eq. (4) the extra pull falls as $e^{-\sqrt{g_N/g_\dagger}}$, so for the Moon around the Earth ($g_N = 2.7 \times 10^{-3}\ {\rm m\,s^{-2}}$) and the Earth around the Sun ($5.9 \times 10^{-3}\ {\rm m\,s^{-2}}$) it is zero to any measurable precision, while the Sun in the Milky Way ($g_N \simeq 2 \times 10^{-10}\ {\rm m\,s^{-2}}$) receives a boost of 1.34, as observed. The square-root form would leave fractional excesses of 2×10^{-8} and 9×10^{-9} in the two Solar System orbits, above what lunar laser ranging and planetary ephemerides allow.

6 Inside our layer: isolated lenses, neighbours and clusters

HMG's current lensing model writes the scale as $g_s = (2c/t)\cos\gamma/\gamma$, with the angle set by a balance parameter $\xi^2 = 1/s^3 + v_H^2/(12v_N^2)$ [5]. Late-type lenses sit on a neighbourhood branch ($s < 1$), early-type and red lenses on a Hubble branch ($s > 1$).

Isolated lenses read the vacuum clock. Setting HMG's g_s equal to the vacuum clock, Eq. (3), gives $s = 0.295$ for the neighbourhood branch, against HMG's fitted 0.329 ± 0.023 for late-type lenses and 0.314 ± 0.023 for blue lenses [5]. With the form of Eq. (4) and one free scale per population, the KiDS-1000 isolated lenses [20] give $1.01\,g_\dagger$ for late-type lenses, $\ln(g/g_\dagger) = +0.01 \pm 0.11$, and $0.85\,g_\dagger$ for blue lenses, -0.16 ± 0.12: both read the vacuum clock within the pass rule of 2σ, with nothing fitted to it.

Early-type lenses carry their neighbours. Early-type and red lenses of the same stellar mass need $2.42\,g_\dagger$ and $2.71\,g_\dagger$. The excess is not one scale per population: the inner and outer halves of each profile differ by 2.8σ (early) and 5.0σ (red), the excess growing from small values close to the lens to a plateau beyond about 0.2 Mpc. Stellar mass-to-light and hot-gas corrections [19] remove 26% of it for early-type lenses and 9% for red lenses. A changed clock would act most where the surroundings are densest, close to the lens; the observed excess grows outward, as the projected mass of correlated neighbours does. With the same environment profile for both readings (the matter correlation function, $r_0 = 5$ Mpc, $\gamma = 1.8$, one free amplitude), added neighbour mass fits early-type and red lenses better than a changed clock by $\Delta\chi^2 = 130$ and 198, with amplitudes 0.47 ± 0.04 and 0.48 ± 0.03; late-type and blue lenses need none. This agrees with the finding that the early-type excess shrinks as the isolation criterion is made stricter [19], and with early-type galaxies living in denser surroundings. In this reading HMG's angle for an isolated structure is fixed by the vacuum clock, and its environment dependence carries the pull of neighbouring structures: in the framework of [6], each neighbour brings its own mass and its own unrelaxed vacuum, the mechanism that fills cluster outskirts one scale up.

Clusters. HMG's cluster model uses a radially varying projection factor $\varepsilon_0^2 = 1/6 + \rho_{\rm enc}/\rho_{\rm vac}$ [4, 5]. Since $\omega = \sqrt{4\pi G\rho}$ for any medium, $\rho_{\rm enc}/\rho_{\rm vac} = (\omega_{\rm matter}/\omega_J)^2$: the square of matter's own rate over the vacuum clock, the same switch that sets where matter relaxes the medium in Eq. (4). With this factor HMG fits the twelve X-COP clusters [21] at a median $n_\sigma = 0.92$ [5].

7 The film: does each layer move with the edge?

The edge grows as $R = ct$. If the vacuum seen as dark energy moved with it, Ω_Λ would be the same at every age and the galactic scale would fall as $1/t$. If the vacuum belongs to its place on S, its layer holds still and the scale stays constant. We take the family

$$\rho_{\rm vac} = \rho_{\rm vac,0}\,(H/H_0)^{2p},$$

in which the layer of the vacuum moves along S as $S_{\rm vac}(t) = S_{\rm vac,0} - p\ln(H/H_0)$, with energy exchanged between vacuum and matter so that the total is conserved; $p = 0$ is a layer that holds still and $p = 1$ a vacuum carried by the edge, with Ω_Λ the same at every age.

Expansion data. Pantheon+, DESI DR2 BAO and Planck distance priors give $p = 0.011 \pm 0.005$: the vacuum was very slightly larger in the past, so the layer drifts toward larger S, where the medium is flatter, as the framework requires [6].

The galactic scale. In the same family, $g_\dagger(z)/g_\dagger(0) = (H/H_0)^p$. The 145 discs at $z = 0.7$–2.5 [22, 23, 24, 25, 26], with distances computed inside the family, give $p = 0.33 \pm 0.20$, consistent with the expansion

data at 1.6σ.

Newton's constant. In the warped geometry the slice's energy content also sets Newton's constant, $G_4 \propto \sigma$ [11, 12]. Lunar laser ranging gives $\dot{G}/G = (7.1 \pm 7.6) \times 10^{-14}\,\mathrm{yr}^{-1}$ [27], so the slice's own energy stays fixed and the motion along S acts through the medium, whose curvature changes with S.

Prediction. The galactic scale stays constant to within 1% out to $z = 2$ ($g_\dagger(1)/g_\dagger(0) = 1.006$, $g_\dagger(2)/g_\dagger(0) = 1.012$), whereas a scale tied to the slice's age, $a \propto 1/t$, would be about twice today's value at $z = 1$. Late-type weak lensing at $z \gtrsim 1$ from Euclid and Rubin decides between them.

Every layer through time. The same question can be put to every layer that carries an interaction. Only dimensionless numbers can be compared across time, so what can be seen is one layer moving relative to another: the particle layers relative to the floor (Newton's constant against particle masses), the strong layer relative to the electron layer (the proton-to-electron mass ratio μ), and the charged medium between the Z mass and the electron (the fine-structure constant α); a shift of every layer together, floor included, is a change of units. If the layers rode with the edge, $S_{\mathrm{layer}}(t) = S_{\mathrm{layer},0} + \ln(t/t_0)$, the particle masses would fall against the Planck mass and $\dot{G}/G$, read through $Gm_p^2/\hbar c$, would be $-2/t_0 = -1.45 \times 10^{-10}\,\mathrm{yr}^{-1}$. Table 5 lists the measurements; the rule, fixed before comparing, is that a layer holds still if its change is within 2σ of zero.

Layers compared	When	Measured change	Source
Particles vs floor ($\dot{G}/G$)	today	$(7.1 \pm 7.6) \times 10^{-14}\,\mathrm{yr}^{-1}$	lunar laser ranging [27]
Strong vs electron ($\Delta\mu/\mu$)	7.5 Gyr ago	$(-1.0 \pm 1.3) \times 10^{-7}$	methanol at $z = 0.89$ [28]
Charged medium ($\dot{\alpha}/\alpha$)	today	$(1.8 \pm 2.5) \times 10^{-19}\,\mathrm{yr}^{-1}$	Yb^+ optical clocks [29]
Charged medium ($\Delta\alpha/\alpha$)	8 Gyr ago	$(1.3 \pm 1.4) \times 10^{-6}$	ESPRESSO, $z = 1.15$ [30]
Charged medium, electron ($\Delta\alpha/\alpha$, $\Delta m_e/m_e$)	13.8 Gyr ago	$(3.6 \pm 3.7) \times 10^{-3}$, $(4 \pm 11) \times 10^{-3}$	CMB [31]
Our vacuum layer (p)	$z \lesssim 2$	0.011 ± 0.005	this section

Table 5: The film checked frame to frame. Every interaction layer holds still within 1σ; our vacuum layer drifts by $p = 0.011$ (2.2σ), far from riding the edge ($p = 1$).

Every interaction layer holds still. Riding the edge is excluded by lunar laser ranging at about 1900σ, and the strong and electron layers have moved relative to each other by less than $|\Delta S| = 2.6 \times 10^{-7}$ in 7.5 Gyr. The layers stay at their places on S while the edge climbs through them (Figure 4); our vacuum layer is the only one where any motion is seen, a slow drift toward larger S.

No phantom crossing. In the family above the vacuum's equation of state is $w = -1 - \frac{2p}{3}\, d\ln H/d\ln a \approx -1 + p\,\Omega_m(a)$, so a layer drifting toward larger S gives $w > -1$ at every redshift ($w_0 = -0.997$ for $p = 0.011$, which refines the $w \approx -0.95$ of the potential in [6]), and a ghost-free motion along S cannot make the vacuum grow [6]. The equation of state never crosses -1. The two-parameter fits to DESI DR2 BAO with the CMB and supernovae prefer a history that crosses -1 near $z \approx 0.4$, at 2.8–4.2σ depending on the supernova sample [14]; reanalyses with recalibrated samples find 1.1–2.3σ [32]. If the crossing is confirmed, the picture of a vacuum layer drifting along S is excluded.

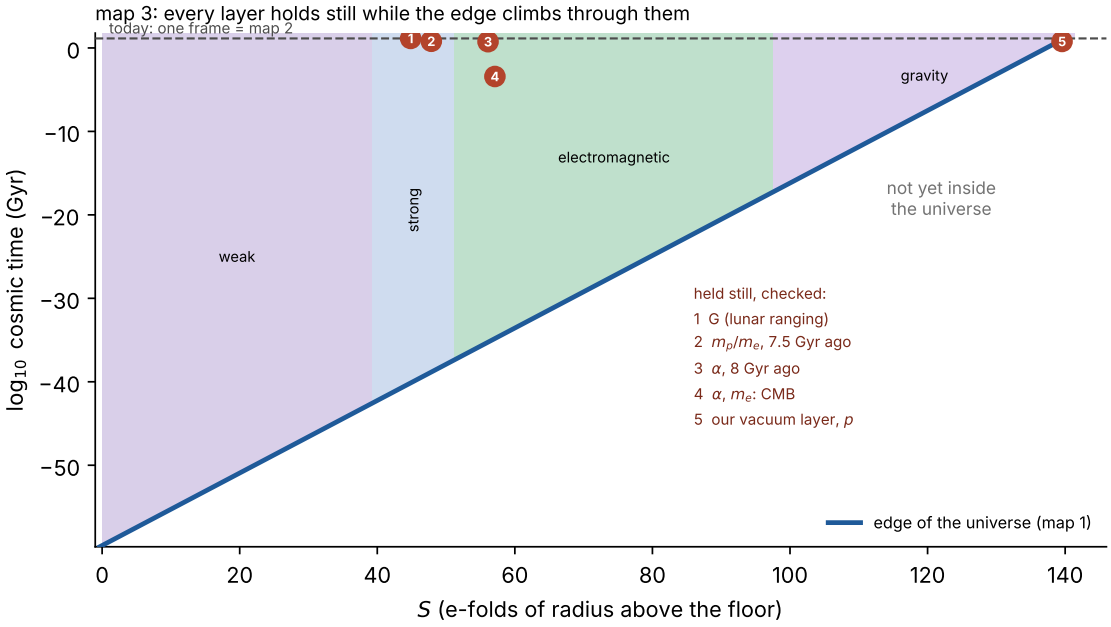


Figure 4: The film: each frame is Figure 2 at one time, bounded by the edge. The bands, what holds the structures at each S, stay where they are as the edge climbs; the numbered points are the checks of Table 5.

8 The edge and the active layer

Along the edge, what is active at each moment is set by temperature, not size: the active layer is the thermal size $\hbar c/kT$ (Figure 5). The edge is the Hubble radius of the standard history; from $z = 2$ to today it is HMG's slice, $R = ct$, within 0.01 e-folds, so every test in Sections 5–7 is made where the two coincide. Into the past the gap between the edge and the active layer closes: nucleosynthesis comes after about 2 minutes and hadrons form at about 10^{-5} s, and the edge, the active layer and the floor meet at the tip of the cone, where the universe is smallest, earliest and hottest at once.

9 Why now: the edge crosses the vacuum's layer

Today the edge, ct_0, sits at $S = 139.90$ and the vacuum's clock scale c/ω_J at $S = 139.92$ ($\omega_J t_0 = 0.986$): the moving edge is crossing the layer that holds still (Figure 1). That is why HMG's values, tied to the edge, and the vacuum clock, tied to the layer, agree today, and why they separate in the past, when the edge was below the layer.

We, as observers, are complex life forms, Earth's life, so we exist only inside Earth's life window ($S = 58$–98), where chemistry and an environment that sustains life are possible. We are built from bound structures at many layers below us: hadrons, atoms, chemistry, stars and the galaxies that hold them. Galaxies form only while matter's own rate beats the medium's clock, that is before the crossing, and the lower layers need several stellar lifetimes to assemble the elements and the chemistry of Earth's life window. We therefore appear near the crossing. This places the arguments of Dicke [33] and Weinberg [34] on one axis.

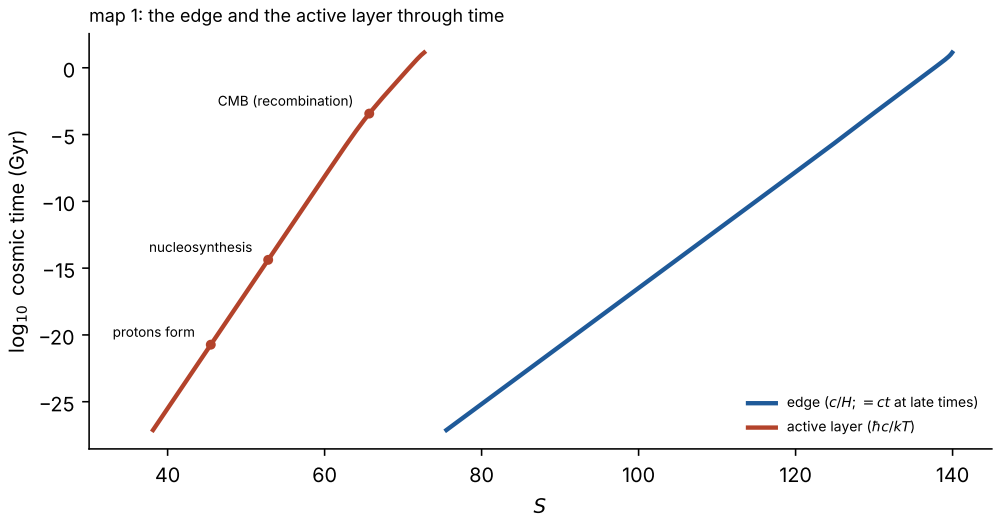


Figure 5: The first map: the edge (blue, the Hubble radius, equal to ct at late times) and the active layer $\hbar c/kT$ (red), with the moments at which protons form, nucleosynthesis happens and the CMB is released.

10 Discussion

What each description brings. HMG brings the slice geometry in full relativity, the projected Ω_Λ that fixes how far above the sign change our slice sits, the environment model and the cluster model. The scale axis brings the second reading of the cone: the layers below the edge, the couplings running across them, the vacuum medium and its curve, the four force layers and the species in the medium whose stack gives the gauge couplings, the vacuum clock that fixes HMG's γ_0 and its isolated-lens baseline, the form of the relation, and the time cut that separates a vacuum carried by the edge from one that holds its place. Neither description is complete without the other: the couplings need the slice and the stack together, and the sheet needs both the edge and the layers.

What is postulated. The paper rests on two readings. The scale axis reads known physics, the running couplings, the vacuum-energy hierarchy and the galactic scale, as positions on one coordinate [6, 7]; HMG reads the expansion as a geometry in which size is a direction. Joining them, the cone's axis is read as scale and each layer is given the clock $t = R/c$: this is the postulate, not a result. The four force layers, each with its own vacuum, and the sign flip at the floor and at our layer are the framework's reading of the axis [6]; they place known physics, they do not derive the coupling values.

What is tested and what is mapped. Where the edge meets our layer every number is measured and none is fitted (Sections 5 and 6), and the time cut is measured at our layer and at every interaction layer (Section 7). Across S, the vacuum is measured at the two ends of the 10^{120}, the floor's quantum-field value and our dark energy, and the couplings at the scales of Table 2; the vacuum of the layers in between follows from the reading of the cone as scale and is not separately measured, since at those layers the condensates are carried by bound structures [6]. The projected Ω_Λ comes on HMG's side from a second-order match of its projected expansion at low redshift [1] and is measured on ours; neither side yet derives the 30% from first principles; the ground-state reading behind Eq. (4) was formulated knowing the observed relation, and its support is that the value and the constancy of the scale are not fitted; the stack's coupling values coincide with Standard Model running, of which Eq. (1) is the reading on the axis, and in it the three gauge couplings do not meet in a single point.

11 Conclusion

HMG's cone can be read as time or as scale, and the two readings are cuts through one sheet. HMG's universe is the sheet's edge, the path of the universe's own size. Across scale today the four interactions are one field read in four layers, each with its own vacuum: HMG's slice formula gives that vacuum, which falls as gravity's coupling by 10^{120} to our layer, the stack of media below each layer gives the gauge couplings, with the strong coupling at the bottom-quark and τ scales and electromagnetism at the electron, and the vacuum changes sign at the floor and at our layer, the two edges of a window on an axis that continues past both. Where the edge meets our layer the vacuum matches the measured dark energy, its Jeans rate gives the galactic scale and HMG's γ_0, isolated lenses read that scale directly, and the excess around early-type lenses is the pull of their neighbours. Along time, every interaction layer holds still, and our layer's vacuum drifts only slowly while the edge moves through it, so the galactic scale is predicted to stay constant to $z = 2$ and the vacuum's equation of state never to cross -1; today the edge crosses that layer.

Data and code

Every number is reproduced by the scripts in the folder `hmg_scale_axis` of `https://github.com/THEREALMIKEONE/Hyperconical-gravity-renormalization-scale-Repository`, with their outputs. They use public data, the test suite of [6] and HMG's own code [5].

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Declarations

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